

AI WINE TESTER

Internship ID

CITS4068

Full Name

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No. of Weeks

4

Project Name

Wine Quality Prediction (AI Wine Tester)

1. Introduction

Wine quality prediction is an important application of Machine Learning in the food and beverage industry. This project aims to predict the quality of wine based on its chemical properties. The model analyzes various features such as acidity, alcohol content, sulphates, density, and pH levels to determine whether the wine is of good or bad quality.

2. Dataset Description

The dataset used for this project is the Wine Quality Dataset. It contains information about different chemical properties of red wine samples.

Features:

- Fixed Acidity
- Volatile Acidity
- Citric Acid
- Residual Sugar
- Chlorides
- Free Sulfur Dioxide
- Total Sulfur Dioxide
- Density
- pH

- Sulphates
- Alcohol
- Quality

The target variable is **Quality**, which represents the wine quality rating.

3. Data Cleaning

The dataset was loaded using the Pandas library and inspected for missing values and inconsistencies.

Steps Performed:

- Loaded dataset into Pandas DataFrame
- Checked dataset dimensions
- Examined dataset information
- Checked for missing values
- Verified data types

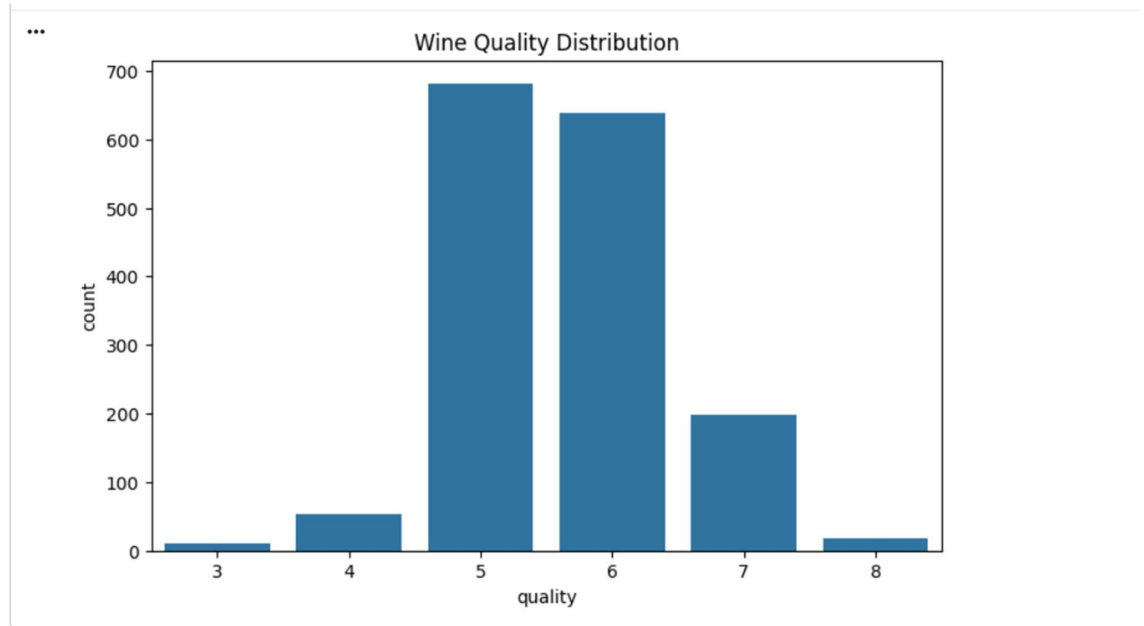
The dataset contained no missing values and was ready for model training.

4. Data Visualization

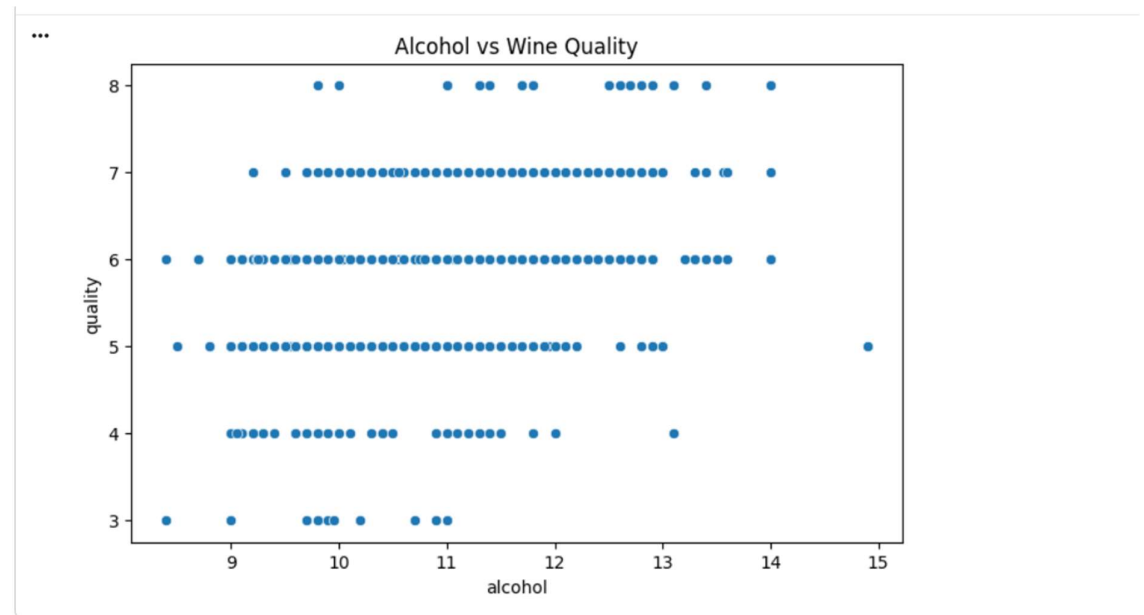
Several visualizations were created to understand the dataset.

Visualizations:

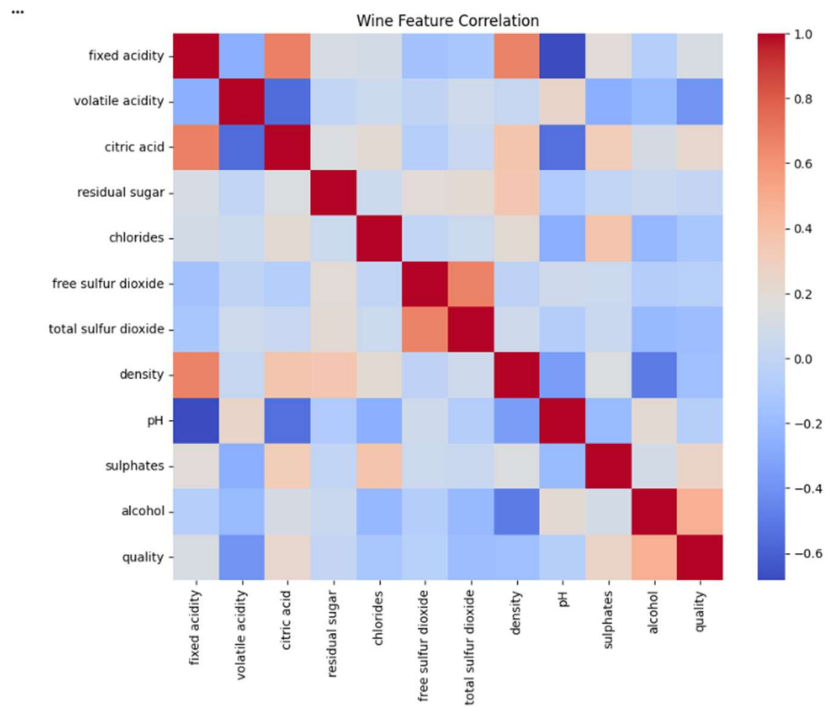
1. Wine Quality Distribution



2. Alcohol vs Wine Quality



3. Correlation Heatmap



These visualizations helped identify relationships between wine features and quality ratings.

Insert Screenshots:

- Dataset Loaded
- Dataset Information
- Wine Quality Distribution
- Alcohol vs Wine Quality
- Correlation Heatmap

5. Model Building

A Random Forest Classifier was used to build the prediction model.

Steps:

- Selected input features
- Created target labels
- Split dataset into training and testing sets
- Trained Random Forest Classifier

- Evaluated model performance

Algorithm Used:

Random Forest Classifier

6. Model Performance

The trained model achieved excellent performance.

Accuracy Achieved:

90.31%

Model Accuracy Output :

```
... Accuracy: 90.31 %
```

7. AI Wine Tester Output

A simple prediction system called **AI Wine Tester** was created.

The model classifies wine into:

- Good Wine (Premium)
- Bad Wine (Average)

Example Output:

AI WINE TESTER

Quality: BAD WINE

Rating: Average

AI Wine Tester Output:

```
... 🍷 AI WINE TESTER
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Quality: BAD WINE
Rating: Average
```

8. Results

The machine learning model successfully predicted wine quality using the provided chemical properties. The project demonstrates how data analysis and machine learning techniques can be used to solve real-world prediction problems.

9. Conclusion

The AI Wine Tester project successfully predicts wine quality with an accuracy of 90.31%. Through data preprocessing, visualization, model training, and evaluation, the project demonstrates the effectiveness of machine learning in quality assessment and decision-making processes.

Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-Learn
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Files Included

- Wine_Quality_Prediction.ipynb
- winequality-red.csv
- wine_quality_prediction.pdf
- Dataset Screenshots
- Output Screenshots
- README.md